Unit 5 Review

Multi-Step Proportional Relationships

Name_____
Date_____
Period_____

Show your work. Circle your final answer.

HOW TO USE THIS. This review covers every learning target from the unit — more than the assessment does. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part.

PART A. Scale Drawings

Lesson 1 · 5 questions

1. A 5 cm side of a figure is 8 cm in its scale copy. Find the scale factor.

2. A 16 cm side of a figure is 4 cm in its scale copy. Find the scale factor.

3. A scale copy is made at k = 1.5. A side of the original is 6 cm. Find the matching side of the copy.

4. A drawing has a 65° angle. Its scale copy is made at k = 2. What is the matching angle in the copy?

- ○ A. 67°
- ○ B. 130°
- ○ C. 32.5°
- ○ D. 65°

5. A map is drawn at 1 cm : 30 km. Two towns are 4.5 cm apart on it. How far apart are they?

PART B. Scale Factor: Perimeter and Area

Lessons 2–3 · 6 questions

6. A figure with perimeter 16 cm is scaled by k = 2.5. Find the copy's perimeter.

7. A figure with area 9 cm^2 is scaled by k = 2.5. Find the copy's area.

8. A 4 cm by 6 cm rectangle is scaled by k = ½.

a. Find the copy's perimeter.

b. Find the copy's area.

9. A figure has area 10 cm^2 and its scale copy has area 160 cm^2. Find k.

10. A figure's perimeter is 12 cm and its copy's perimeter is 36 cm. What is the area ratio, copy to original?

11. Sam says tripling every side of a rectangle triples its area. Explain why not, with a counterexample.

PART C. Real-World Scale Drawings

Lesson 4 · 4 questions

12. A plan is drawn at ½ in : 1 ft. A wall is 7.5 in long on it. Find the real length.

13. A plan is drawn at 1 in : 6 ft. A real wall is 21 ft long. Find its length on the plan.

14. A model boat is built at 1 : 25 and is 32 cm long. Find the real boat's length in metres.

15. Every part below is about one floor plan drawn at ¼ in : 1 ft, on which a room is 3 in by 4 in. 1 ft = 0.305 m.

a. Find the real length of the wall drawn 3 in long.

b. Find the room's real area.

c. Write the scale factor from the drawing to the room as a single number.

d. Find the room's real area in square metres.

PART D. Converting Length Across Systems; Currency

Lessons 6–7 · 6 questions

16. 1 in = 2.54 cm. How many centimetres is 14 in?

17. 1 cm = 0.39 in. How many inches is 120 cm?

18. 1 km = 0.62 mi. How many miles is 30 km?

19. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimetres is 6 yd? Estimate first.

20. Rate on 3 Nov: 1 US dollar = 0.92 euros.

a. How many euros is $150?

b. How many US dollars is 69 euros?

21. 1 in = 2.54 cm; 1 cm = 0.39 in. Lee converts 40 cm to inches by multiplying by 2.54 and gets 101.6 in. Name the error and fix it.

PART E. Converting Mass, Area and Volume

Lessons 8–9 · 5 questions

22. 1 lb = 0.454 kg. How many kilograms is 80 lb?

23. 1 L = 0.264 gal. How many gallons is 40 L?

24. 1 ft = 12 in. How many square inches is 8 ft^2?

25. 1 c = 8 fl oz and 1 c = 0.24 L. How many litres is 48 fl oz?

26. 1 m = 100 cm. A desk is 1.5 m^2 in area. How many square centimetres is that?

- ○ A. 300 cm^2
- ○ B. 150 cm^2
- ○ C. 15,000 cm^2
- ○ D. 1,500 cm^2

PART F. Multi-Step Proportion Problems

Lessons 10–11 · 4 questions

27. A car does 29 miles per gallon and its tank holds 12 gal.

a. How far does it go on a full tank?

b. It starts an 800-mi trip with a full tank. How many tanks of gas does the trip take?

28. A recipe for 3 servings uses 1¼ cups of rice.

a. How much rice for 12 servings?

b. With 10 cups on hand, how many servings can be made?

29. 1 km = 0.62 mi. A train does 110 km/h; a bus does 65 mi/h. Which is faster? Show both speeds in one unit.

30. Paint is mixed 1 : 2 : 5. A batch uses 10 gal of the third colour. How many gallons is the whole batch?

GRADE 7 ACCELERATED · UNIT 5

Unit 5 Review

Multi-Step Proportional Relationships **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

HOW TO USE THIS. This review covers every learning target from the unit — more than the assessment does. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part.

PART A. Scale Drawings

Lesson 1 · MA.7.GR.1.5 · 5 questions

1. A 5 cm side of a figure is 8 cm in its scale copy. Find the scale factor.

k = 1.6

copy ÷ original

2. A 16 cm side of a figure is 4 cm in its scale copy. Find the scale factor.

k = 0.25

the copy is smaller, so k is under 1

3. A scale copy is made at k = 1.5. A side of the original is 6 cm. Find the matching side of the copy.

9 cm

4. A drawing has a 65° angle. Its scale copy is made at k = 2. What is the matching angle in the copy?

- ○ A. 67°
- ○ B. 130°
- ○ C. 32.5°
- ● **D. 65°**

D

5. A map is drawn at 1 cm : 30 km. Two towns are 4.5 cm apart on it. How far apart are they?

135 km

drawing × rate

PART B. Scale Factor: Perimeter and Area

Lessons 2–3 · MA.7.GR.1.5 · 6 questions

6. A figure with perimeter 16 cm is scaled by k = 2.5. Find the copy's perimeter.

40 cm

perimeter × k

7. A figure with area 9 cm^2 is scaled by k = 2.5. Find the copy's area.

56.25 cm^2

area × k^2 = 9 × 6.25

8. A 4 cm by 6 cm rectangle is scaled by k = ½.

a. Find the copy's perimeter.

10 cm

20 × ½

b. Find the copy's area.

6 cm^2

24 × ¼

9. A figure has area 10 cm^2 and its scale copy has area 160 cm^2. Find k.

k = 4

160 ÷ 10 = 16 = k^2

10. A figure's perimeter is 12 cm and its copy's perimeter is 36 cm. What is the area ratio, copy to original?

9

k = 3 from the perimeters; the area ratio is k^2

11. Sam says tripling every side of a rectangle triples its area. Explain why not, with a counterexample.

A 2 by 5 (area 10) tripled is 6 by 15 (area 90): nine times, because area is two lengths and each is tripled.

the reason and the numbers

PART C. Real-World Scale Drawings

Lesson 4 · MA.7.GR.1.5 · 4 questions

12. A plan is drawn at ½ in : 1 ft. A wall is 7.5 in long on it. Find the real length.

15 ft

1 in : 2 ft; 7.5 × 2

13. A plan is drawn at 1 in : 6 ft. A real wall is 21 ft long. Find its length on the plan.

3.5 in

real ÷ rate

14. A model boat is built at 1 : 25 and is 32 cm long. Find the real boat's length in metres.

8 m

32 × 25 = 800 cm

15. **SSDD.** Every part below is about one floor plan drawn at ¼ in : 1 ft, on which a room is 3 in by 4 in. 1 ft = 0.305 m. Same surface, four different questions: a length, an area with the rate used twice, the scale factor as one number, and a conversion — the four moves of the unit on one room, so that a student learns to read the question and not the picture.

a. Find the real length of the wall drawn 3 in long.

12 ft

1 in : 4 ft; 3 × 4

b. Find the room's real area.

192 ft²

12 by 16; the rate used twice

c. Write the scale factor from the drawing to the room as a single number.

48

1 in : 4 ft is 1 in : 48 in

d. Find the room's real area in square metres.

About 17.86 m²

3.66 m by 4.88 m = 17.8608 m²; or 192 × 0.305²

PART D. Converting Length Across Systems; Currency *Lessons 6–7 · MA.7.AR.3.3 · 6 questions*

16. 1 in = 2.54 cm. How many centimetres is 14 in?

35.56 cm

17. 1 cm = 0.39 in. How many inches is 120 cm?

46.8 in (47.2 by ÷ 2.54)

fewer inches than centimetres

18. 1 km = 0.62 mi. How many miles is 30 km?

18.6 mi

19. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimetres is 6 yd? Estimate first.

About 540; exactly 548.64 cm

216 in × 2.54

20. Rate on 3 Nov: 1 US dollar = 0.92 euros.

a. How many euros is $150?

138 euros

× the rate

b. How many US dollars is 69 euros?

$75

÷ the rate

21. 1 in = 2.54 cm; 1 cm = 0.39 in. Lee converts 40 cm to inches by multiplying by 2.54 and gets 101.6 in. Name the error and fix it.

Lee used the ratio whose 1 is inches on centimetres. Inches are bigger, so there are fewer: 40 × 0.39 = 15.6 in (or 40 ÷ 2.54 = 15.7).

the error named and the estimate

PART E. Converting Mass, Area and Volume

Lessons 8–9 · MA.7.AR.3.3 · 5 questions

22. 1 lb = 0.454 kg. How many kilograms is 80 lb?

36.32 kg

(80 ÷ 2.204 = 36.30 by the other ratio)

23. 1 L = 0.264 gal. How many gallons is 40 L?

10.56 gal

24. 1 ft = 12 in. How many square inches is 8 ft^2?

1,152 in^2

8 × 144: the ratio squared

25. 1 c = 8 fl oz and 1 c = 0.24 L. How many litres is 48 fl oz?

1.44 L

6 c × 0.24

26. 1 m = 100 cm. A desk is 1.5 m^2 in area. How many square centimetres is that?

- ○ A. 300 cm^2
- ○ B. 150 cm^2
- ● **C. 15,000 cm^2**
- ○ D. 1,500 cm^2

C

PART F. Multi-Step Proportion Problems

Lessons 10–11 · MA.7.AR.4.5 · 4 questions

27. A car does 29 miles per gallon and its tank holds 12 gal.

a. How far does it go on a full tank?

348 mi

b. It starts an 800-mi trip with a full tank. How many tanks of gas does the trip take?

3 tanks (800 ÷ 348 ≈ 2.3)

a part of a tank is still a tank

28. A recipe for 3 servings uses 1¼ cups of rice.

a. How much rice for 12 servings?

5 cups

k = 4

b. With 10 cups on hand, how many servings can be made?

24 servings

8 recipes × 3

29. 1 km = 0.62 mi. A train does 110 km/h; a bus does 65 mi/h. Which is faster? Show both speeds in one unit.

The train: 110 km/h = 68.2 mi/h > 65 mi/h

same unit first

30. Paint is mixed 1 : 2 : 5. A batch uses 10 gal of the third colour. How many gallons is the whole batch?

16 gal

k = 2; 8 parts × 2

GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

Multi-Step Proportional Relationships

Form A · 24 points · two class periods

Name_____
Date_____
Period_____

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors

10 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 4 cm side of a figure is 10 cm in its scale copy. Find the scale factor. *1 pt*

2. A 15 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

3. A drawing has a 35° angle. A scale copy is made at k = 3. What is the matching angle in the copy? *1 pt*

- ○ A. 105°
- ○ B. 35°
- ○ C. 11.67°
- ○ D. 38°

4. A 3 cm by 4 cm rectangle is scaled by k = 3.

a. Find the copy's perimeter. *1 pt*

b. Find the copy's area. *1 pt*

5. A figure has area 8 cm^2. Its scale copy has area 72 cm^2. Find k, and explain why k is not 9. *1 pt*

6. A floor plan is drawn at ¼ in : 1 ft. A room is 2 in by 3.5 in on the plan.

a. Find the real length of the wall drawn 2 in long. *1 pt*

b. Find the room's real area. *1 pt*

7. A plan is drawn at 1 in : 4 ft. A real wall is 18 ft long. How long is it on the plan? *1 pt*

8. A plan is drawn at 1 in : 5 ft. A room is 2 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square metres. *1 pt*

Part 2. Converting Across Systems *8 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimetres is 9 in? *1 pt*

10. 1 cm = 0.39 in. How many inches is 70 cm? *1 pt*

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimetres is 4 yd? *1 pt*

12. 1 kg = 2.204 lb. How many pounds is 20 kg? *1 pt*

13. 1 gal = 3.785 L. How many litres is 5 gal? *1 pt*

14. 1 yd = 3 ft. A rug is 4 yd^2 in area. How many square feet is that? *1 pt*

- ○ A. 12 ft^2
- ○ B. 9 ft^2
- ○ C. 36 ft^2
- ○ D. 24 ft^2

15. Rate on 5 Nov: 1 US dollar = 0.92 euros.

a. How many euros is $60? *1 pt*

b. How many US dollars is 23 euros? *1 pt*

Part 3. Multi-Step Proportion Problems *6 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car does 26 miles per gallon and its tank holds 15 gal.

a. How far does it go on a full tank? *1 pt*

b. It starts a 1,000-mi trip with a full tank. How many tanks of gas does the trip take? *1 pt*

17. A recipe for 4 servings uses 1½ cups of rice.

a. How much rice for 10 servings? *1 pt*

b. With 6 cups on hand, how many servings can be made? *1 pt*

18. Juice is mixed orange : mango : water = 3 : 2 : 4. A batch uses 9 cups of orange. How many cups is the whole batch? *1 pt*

19. 1 mi = 1.61 km. A 200-km drive. Car A averages 40 mi/h; car B averages 60 km/h. Which car gets there first, and by about how long? Explain. *1 pt*

GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

Multi-Step Proportional Relationships

Form A · 24 points · two class periods **ANSWER KEY**

Name
Date
Period

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors

10 points MA.7.GR.1.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 4 cm side of a figure is 10 cm in its scale copy. Find the scale factor. *1 pt*

k = 2.5

copy ÷ original

Scoring: 1 point.

2. A 15 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

k = 0.4

copy ÷ original; the copy is smaller, so k is under 1 — 2.5, 4 and 2 are the proportion backwards

Scoring: 1 point.

3. A drawing has a 35° angle. A scale copy is made at k = 3. What is the matching angle in the copy? *1 pt*

- ○ A. 105°
- ● **B. 35°**
- ○ C. 11.67°
- ○ D. 38°

B

Lengths scale; angles do not.

Scoring: 1 point.

4. A 3 cm by 4 cm rectangle is scaled by k = 3.

a. Find the copy's perimeter. *1 pt*

42 cm

perimeter × k

b. Find the copy's area. *1 pt*

108 cm²

area × k² — 36, 40 and 30 multiplied the area by k once

Scoring: a 1 · b 1 = 2 points.

5. A figure has area 8 cm². Its scale copy has area 72 cm². Find k, and explain why k is not 9. *1 pt*

k = 3. The area ratio is k², because area is two lengths each multiplied by k; 9 is k², so k is its square root.

The point needs k AND the reason (two lengths, so the ratio is k²). 'k = 3' alone is half the item; 'because you square root it' with no reason why is the rule without the reason.

Scoring: 1 point.

6. A floor plan is drawn at ¼ in : 1 ft. A room is 2 in by 3.5 in on the plan.

a. Find the real length of the wall drawn 2 in long. *1 pt*

8 ft

the scale per 1 inch, then × — 1 in : 4 ft, 1 in : 2 ft, 1 in : 5 ft

b. Find the room's real area. *1 pt*

112 ft²

real dimensions first: 8 by 14, 6 by 10, 10 by 15 — the drawing's area × the rate once is the named error

Scoring: a 1 · b 1 = 2 points.

7. A plan is drawn at 1 in : 4 ft. A real wall is 18 ft long. How long is it on the plan? *1 pt*

4.5 in

real ÷ rate

Scoring: 1 point.

8. A plan is drawn at 1 in : 5 ft. A room is 2 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square metres. *1 pt*

About 9.30 m^2 (3.05 m by 3.05 m)

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. 10 ft by 10 ft; 3.05 × 3.05 = 9.3025. The same number by the other route: 100 ft^2 × 0.305^2 = 9.3025. 30.5 (100 × 0.305) used the ratio once on an area. A plan's area asked in metric appears on no review and in no question bank; its type was 5.10 board 8.

Scoring: 1 point.

Part 2. Converting Across Systems

8 points MA.7.AR.3.3

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimetres is 9 in? *1 pt*

22.86 cm

× the ratio whose 1 is inches

Scoring: 1 point.

10. 1 cm = 0.39 in. How many inches is 70 cm? *1 pt*

27.3 in (27.6 by ÷ 2.54)

× 0.39; the sheet's other ratio, ÷ 2.54, earns the point too when the work shows it. × 2.54 (177.8, 228.6, 279.4) is the inverted ratio.

Scoring: 1 point.

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimetres is 4 yd? *1 pt*

365.76 cm

a chain: 144 in, 180 in, 54 in, then × 2.54. The estimate (a yard is nearly a metre) catches 10.16, 12.7 and 3.81.

Scoring: 1 point.

12. 1 kg = 2.204 lb. How many pounds is 20 kg? *1 pt*

44.08 lb

× the ratio whose 1 is the unit you have; the estimate says more pounds than kilograms

Scoring: 1 point.

13. 1 gal = 3.785 L. How many litres is 5 gal? *1 pt*

18.925 L

more litres than gallons

Scoring: 1 point.

14. 1 yd = 3 ft. A rug is 4 yd^2 in area. How many square feet is that? *1 pt*

- ○ A. 12 ft^2
- ○ B. 9 ft^2
- ● **C. 36 ft^2**
- ○ D. 24 ft^2

C

Scoring: 1 point.

15. Rate on 5 Nov: 1 US dollar = 0.92 euros.

a. How many euros is $60? *1 pt*

55.20 euros

× the rate

b. How many US dollars is 23 euros? *1 pt*

$25

÷ the rate; 21.16, 72.90 and 87.12 multiplied both ways

Scoring: a 1 · b 1 = 2 points.

Part 3. Multi-Step Proportion Problems

6 points MA.7.AR.4.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car does 26 miles per gallon and its tank holds 15 gal.

a. How far does it go on a full tank? *1 pt*

390 mi

the rate × the gallons

b. It starts a 1,000-mi trip with a full tank. How many tanks of gas does the trip take? *1 pt*

3 tanks (1,000 ÷ 390 ≈ 2.56)

the part of a tank is still a tank; follow-through on part (a)

Scoring: a 1 · b 1 = 2 points.

17. A recipe for 4 servings uses 1½ cups of rice.

a. How much rice for 10 servings? *1 pt*

3¾ cups

k from the yields (2.5, 1.5, 3), then the ingredient

b. With 6 cups on hand, how many servings can be made? *1 pt*

16 servings

recipes the pantry makes (4, 4, 5) × the yield; 15, 24 and 25 scaled the yield by the cups

Scoring: a 1 · b 1 = 2 points.

18. Juice is mixed orange : mango : water = 3 : 2 : 4. A batch uses 9 cups of orange. How many cups is the whole batch? *1 pt*

27 cups

k from the part you know (3, 4, 4), then the whole (9, 8, 6 parts); 12, 16 and 8 found one other part and stopped

Scoring: 1 point.

19. 1 mi = 1.61 km. A 200-km drive. Car A averages 40 mi/h; car B averages 60 km/h. Which car gets there first, and by about how long? Explain. *1 pt*

Car A: 40 mi/h = 64.4 km/h > 60 km/h, so about 3.11 h against 3.33 h — about 13 minutes sooner.

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. The point needs the speeds in one unit, the choice, and the times (or the gap). 'Car B, 60 > 40' compared two units. A decision from two rates in different units appears on no review and in no question bank; its type was 5.10 board 9.

Scoring: 1 point.

Score Tracker

Benchmark	Questions	Possible	Earned
MA.7.GR.1.5	1, 2, 3, 4, 5, 6, 7, 8	10	
MA.7.AR.3.3	9, 10, 11, 12, 13, 14, 15	8	
MA.7.AR.4.5	16, 17, 18, 19	6	
Total		**24**	

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.

GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

Multi-Step Proportional Relationships

Form A · 24 points · two class periods **WORKED ANSWERS**

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors *10 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 4 cm side of a figure is 10 cm in its scale copy. Find the scale factor. *1 pt*

k = 2.5

copy ÷ original

Scoring: 1 point.

2. A 15 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

k = 0.4

copy ÷ original; the copy is smaller, so k is under 1 — 2.5, 4 and 2 are the proportion backwards

Scoring: 1 point.

3. A drawing has a 35° angle. A scale copy is made at k = 3. What is the matching angle in the copy? *1 pt*

- ○ A. 105°
- ● **B. 35°**
- ○ C. 11.67°
- ○ D. 38°

B

Lengths scale; angles do not.

Scoring: 1 point.

4. A 3 cm by 4 cm rectangle is scaled by k = 3.

a. Find the copy's perimeter. *1 pt*

42 cm

perimeter × k

b. Find the copy's area. *1 pt*

108 cm²

area × k² — 36, 40 and 30 multiplied the area by k once

Scoring: a 1 · b 1 = 2 points.

5. A figure has area 8 cm². Its scale copy has area 72 cm². Find k, and explain why k is not 9. *1 pt*

k = 3. The area ratio is k², because area is two lengths each multiplied by k; 9 is k², so k is its square root.

The point needs k AND the reason (two lengths, so the ratio is k²). 'k = 3' alone is half the item; 'because you square root it' with no reason why is the rule without the reason.

Scoring: 1 point.

6. A floor plan is drawn at ¼ in : 1 ft. A room is 2 in by 3.5 in on the plan.

a. Find the real length of the wall drawn 2 in long. *1 pt*

8 ft

the scale per 1 inch, then × — 1 in : 4 ft, 1 in : 2 ft, 1 in : 5 ft

b. Find the room's real area. *1 pt*

112 ft²

real dimensions first: 8 by 14, 6 by 10, 10 by 15 — the drawing's area × the rate once is the named error

Scoring: a 1 · b 1 = 2 points.

7. A plan is drawn at 1 in : 4 ft. A real wall is 18 ft long. How long is it on the plan? *1 pt*

4.5 in

real ÷ rate

Scoring: 1 point.

8. A plan is drawn at 1 in : 5 ft. A room is 2 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square metres. *1 pt*

About 9.30 m^2 (3.05 m by 3.05 m)

10 ft by 10 ft; 3.05 × 3.05 = 9.3025. The same number by the other route: 100 ft^2 × 0.305^2 = 9.3025. 30.5 (100 × 0.305) used the ratio once on an area. A plan's area asked in metric appears on no review and in no question bank; its type was 5.10 board 8.

Scoring: 1 point.

Part 2. Converting Across Systems *8 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimetres is 9 in? *1 pt*

22.86 cm

× the ratio whose 1 is inches

Scoring: 1 point.

10. 1 cm = 0.39 in. How many inches is 70 cm? *1 pt*

27.3 in (27.6 by ÷ 2.54)

× 0.39; the sheet's other ratio, ÷ 2.54, earns the point too when the work shows it. × 2.54 (177.8, 228.6, 279.4) is the inverted ratio.

Scoring: 1 point.

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimetres is 4 yd? *1 pt*

365.76 cm

a chain: 144 in, 180 in, 54 in, then × 2.54. The estimate (a yard is nearly a metre) catches 10.16, 12.7 and 3.81.

Scoring: 1 point.

12. 1 kg = 2.204 lb. How many pounds is 20 kg? *1 pt*

44.08 lb

× the ratio whose 1 is the unit you have; the estimate says more pounds than kilograms

Scoring: 1 point.

13. 1 gal = 3.785 L. How many litres is 5 gal? *1 pt*

18.925 L

more litres than gallons

Scoring: 1 point.

14. 1 yd = 3 ft. A rug is 4 yd^2 in area. How many square feet is that? *1 pt*

- ○ A. 12 ft^2
- ○ B. 9 ft^2
- ● **C. 36 ft^2**
- ○ D. 24 ft^2

C

Scoring: 1 point.

15. Rate on 5 Nov: 1 US dollar = 0.92 euros.

a. How many euros is $60? *1 pt*

55.20 euros

× the rate

b. How many US dollars is 23 euros? *1 pt*

$25

÷ the rate; 21.16, 72.90 and 87.12 multiplied both ways

Scoring: a 1 · b 1 = 2 points.

Part 3. Multi-Step Proportion Problems *6 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car does 26 miles per gallon and its tank holds 15 gal.

a. How far does it go on a full tank? *1 pt*

390 mi

the rate × the gallons

b. It starts a 1,000-mi trip with a full tank. How many tanks of gas does the trip take? *1 pt*

3 tanks (1,000 ÷ 390 ≈ 2.56)

the part of a tank is still a tank; follow-through on part (a)

Scoring: a 1 · b 1 = 2 points.

17. A recipe for 4 servings uses 1½ cups of rice.

a. How much rice for 10 servings? *1 pt*

3¾ cups

k from the yields (2.5, 1.5, 3), then the ingredient

b. With 6 cups on hand, how many servings can be made? *1 pt*

16 servings

recipes the pantry makes (4, 4, 5) × the yield; 15, 24 and 25 scaled the yield by the cups

Scoring: a 1 · b 1 = 2 points.

18. Juice is mixed orange : mango : water = 3 : 2 : 4. A batch uses 9 cups of orange. How many cups is the whole batch? *1 pt*

27 cups

k from the part you know (3, 4, 4), then the whole (9, 8, 6 parts); 12, 16 and 8 found one other part and stopped

Scoring: 1 point.

19. 1 mi = 1.61 km. A 200-km drive. Car A averages 40 mi/h; car B averages 60 km/h. Which car gets there first, and by about how long? Explain. *1 pt*

Car A: 40 mi/h = 64.4 km/h > 60 km/h, so about 3.11 h against 3.33 h — about 13 minutes sooner.

The point needs the speeds in one unit, the choice, and the times (or the gap). 'Car B, 60 > 40' compared two units. A decision from two rates in different units appears on no review and in no question bank; its type was 5.10 board 9.

Scoring: 1 point.

GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

Multi-Step Proportional Relationships

Form B · 24 points · two class periods

Name_____
Date_____
Period_____

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors

10 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 6 cm side of a figure is 9 cm in its scale copy. Find the scale factor. *1 pt*

2. A 20 cm side of a figure is 5 cm in its scale copy. Find the scale factor. *1 pt*

3. A drawing has a 50° angle. A scale copy is made at k = 2. What is the matching angle in the copy? *1 pt*

- ○ A. 100°
- ○ B. 25°
- ○ C. 50°
- ○ D. 52°

4. A 2 cm by 5 cm rectangle is scaled by k = 4.

a. Find the copy's perimeter. *1 pt*

b. Find the copy's area. *1 pt*

5. A figure has area 5 cm^2. Its scale copy has area 80 cm^2. Find k, and explain why k is not 16. *1 pt*

6. A floor plan is drawn at ½ in : 1 ft. A room is 3 in by 5 in on the plan.

a. Find the real length of the wall drawn 3 in long. *1 pt*

b. Find the room's real area. *1 pt*

7. A plan is drawn at ¼ in : 1 ft. A real wall is 22 ft long. How long is it on the plan? *1 pt*

8. A plan is drawn at 1 in : 4 ft. A room is 1.5 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square metres. *1 pt*

Part 2. Converting Across Systems *8 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimetres is 11 in? *1 pt*

10. 1 cm = 0.39 in. How many inches is 90 cm? *1 pt*

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimetres is 5 yd? *1 pt*

12. 1 lb = 0.454 kg. How many kilograms is 60 lb? *1 pt*

13. 1 gal = 3.785 L. How many litres is 9 gal? *1 pt*

14. 1 ft = 12 in. A tile is 7 ft^2 in area. How many square inches is that? *1 pt*

- ○ A. 1,008 in^2
- ○ B. 84 in^2
- ○ C. 144 in^2
- ○ D. 168 in^2

15. Rate on 5 Nov: 1 US dollar = 1.35 Canadian dollars.

a. How many Canadian dollars is $70? *1 pt*

b. How many US dollars is 54 Canadian dollars? *1 pt*

Part 3. Multi-Step Proportion Problems *6 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car does 22 miles per gallon and its tank holds 20 gal.

a. How far does it go on a full tank? *1 pt*

b. It starts a 1,500-mi trip with a full tank. How many tanks of gas does the trip take? *1 pt*

17. A recipe for 6 servings uses 2 cups of rice.

a. How much rice for 9 servings? *1 pt*

b. With 8 cups on hand, how many servings can be made? *1 pt*

18. Paint is mixed 1 : 3 : 4. A batch uses 12 gal of the second colour. How many gallons is the whole batch? *1 pt*

19. 1 mi = 1.61 km. A 450-km drive. Truck A averages 55 mi/h; truck B averages 90 km/h. Which truck gets there first, and by about how long? Explain. *1 pt*

GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

Multi-Step Proportional Relationships

Form B · 24 points · two class periods **ANSWER KEY**

Name
Date
Period

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors

10 points MA.7.GR.1.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 6 cm side of a figure is 9 cm in its scale copy. Find the scale factor. *1 pt*

k = 1.5

copy ÷ original

Scoring: 1 point.

2. A 20 cm side of a figure is 5 cm in its scale copy. Find the scale factor. *1 pt*

k = 0.25

copy ÷ original; the copy is smaller, so k is under 1 — 2.5, 4 and 2 are the proportion backwards

Scoring: 1 point.

3. A drawing has a 50° angle. A scale copy is made at k = 2. What is the matching angle in the copy? *1 pt*

- ○ A. 100°
- ○ B. 25°
- ● **C. 50°**
- ○ D. 52°

C

Lengths scale; angles do not.

Scoring: 1 point.

4. A 2 cm by 5 cm rectangle is scaled by k = 4.

a. Find the copy's perimeter. *1 pt*

56 cm

perimeter × k

b. Find the copy's area. *1 pt*

160 cm²

area × k² — 36, 40 and 30 multiplied the area by k once

Scoring: a 1 · b 1 = 2 points.

5. A figure has area 5 cm². Its scale copy has area 80 cm². Find k, and explain why k is not 16. *1 pt*

k = 4. The area ratio is k², because area is two lengths each multiplied by k; 16 is k², so k is its square root.

The point needs k AND the reason (two lengths, so the ratio is k²). 'k = 3' alone is half the item; 'because you square root it' with no reason why is the rule without the reason.

Scoring: 1 point.

6. A floor plan is drawn at ½ in : 1 ft. A room is 3 in by 5 in on the plan.

a. Find the real length of the wall drawn 3 in long. *1 pt*

6 ft

the scale per 1 inch, then × — 1 in : 4 ft, 1 in : 2 ft, 1 in : 5 ft

b. Find the room's real area. *1 pt*

60 ft²

real dimensions first: 8 by 14, 6 by 10, 10 by 15 — the drawing's area × the rate once is the named error

Scoring: a 1 · b 1 = 2 points.

7. A plan is drawn at ¼ in : 1 ft. A real wall is 22 ft long. How long is it on the plan? *1 pt*

5.5 in

real ÷ rate

Scoring: 1 point.

8. A plan is drawn at 1 in : 4 ft. A room is 1.5 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square metres. *1 pt*

About 4.47 m^2 (1.83 m by 2.44 m)

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. 6 ft by 8 ft; 1.83 × 2.44 = 4.4652. The same number by the other route: 48 ft^2 × 0.305^2 = 4.4652. 14.64 (48 × 0.305) used the ratio once on an area. A plan's area asked in metric appears on no review and in no question bank; its type was 5.10 board 8.

Scoring: 1 point.

Part 2. Converting Across Systems

8 points MA.7.AR.3.3

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimetres is 11 in? *1 pt*

27.94 cm

× the ratio whose 1 is inches

Scoring: 1 point.

10. 1 cm = 0.39 in. How many inches is 90 cm? *1 pt*

35.1 in (35.4 by ÷ 2.54)

× 0.39; the sheet's other ratio, ÷ 2.54, earns the point too when the work shows it. × 2.54 (177.8, 228.6, 279.4) is the inverted ratio.

Scoring: 1 point.

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimetres is 5 yd? *1 pt*

457.2 cm

a chain: 144 in, 180 in, 54 in, then × 2.54. The estimate (a yard is nearly a metre) catches 10.16, 12.7 and 3.81.

Scoring: 1 point.

12. 1 lb = 0.454 kg. How many kilograms is 60 lb? *1 pt*

27.24 kg

× the ratio whose 1 is the unit you have; the estimate says more pounds than kilograms

Scoring: 1 point.

13. 1 gal = 3.785 L. How many litres is 9 gal? *1 pt*

34.065 L

more litres than gallons

Scoring: 1 point.

14. 1 ft = 12 in. A tile is 7 ft^2 in area. How many square inches is that? *1 pt*

- ● **A. 1,008 in^2**
- ○ B. 84 in^2
- ○ C. 144 in^2
- ○ D. 168 in^2

A

Scoring: 1 point.

15. Rate on 5 Nov: 1 US dollar = 1.35 Canadian dollars.

a. How many Canadian dollars is $70? *1 pt*

94.50 Canadian dollars

× the rate

b. How many US dollars is 54 Canadian dollars? *1 pt*

$40

÷ the rate; 21.16, 72.90 and 87.12 multiplied both ways

Scoring: a 1 · b 1 = 2 points.

Part 3. Multi-Step Proportion Problems

6 points MA.7.AR.4.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car does 22 miles per gallon and its tank holds 20 gal.

a. How far does it go on a full tank? *1 pt*

440 mi

the rate × the gallons

b. It starts a 1,500-mi trip with a full tank. How many tanks of gas does the trip take? *1 pt*

4 tanks (1,500 ÷ 440 ≈ 3.41)

the part of a tank is still a tank; follow-through on part (a)

Scoring: a 1 · b 1 = 2 points.

17. A recipe for 6 servings uses 2 cups of rice.

a. How much rice for 9 servings? *1 pt*

3 cups

k from the yields (2.5, 1.5, 3), then the ingredient

b. With 8 cups on hand, how many servings can be made? *1 pt*

24 servings

recipes the pantry makes (4, 4, 5) × the yield; 15, 24 and 25 scaled the yield by the cups

Scoring: a 1 · b 1 = 2 points.

18. Paint is mixed 1 : 3 : 4. A batch uses 12 gal of the second colour. How many gallons is the whole batch? *1 pt*

32 gal

k from the part you know (3, 4, 4), then the whole (9, 8, 6 parts); 12, 16 and 8 found one other part and stopped

Scoring: 1 point.

19. 1 mi = 1.61 km. A 450-km drive. Truck A averages 55 mi/h; truck B averages 90 km/h. Which truck gets there first, and by about how long? Explain. *1 pt*

Truck B: 55 mi/h = 88.55 km/h < 90 km/h, so 5 h against about 5.08 h — about 5 minutes sooner.

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. The point needs the speeds in one unit, the choice, and the times (or the gap). 'Truck B, 90 > 55' is right by accident — no credit without the conversion. The surface appears on no review and in no question bank; its type was 5.10 board 9.

Scoring: 1 point.

Score Tracker

Benchmark	Questions	Possible	Earned
MA.7.GR.1.5	1, 2, 3, 4, 5, 6, 7, 8	10	
MA.7.AR.3.3	9, 10, 11, 12, 13, 14, 15	8	
MA.7.AR.4.5	16, 17, 18, 19	6	
Total		**24**	

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.

GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

Multi-Step Proportional Relationships

Form B · 24 points · two class periods **WORKED ANSWERS**

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors *10 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 6 cm side of a figure is 9 cm in its scale copy. Find the scale factor. *1 pt*

k = 1.5

copy ÷ original

Scoring: 1 point.

2. A 20 cm side of a figure is 5 cm in its scale copy. Find the scale factor. *1 pt*

k = 0.25

copy ÷ original; the copy is smaller, so k is under 1 — 2.5, 4 and 2 are the proportion backwards

Scoring: 1 point.

3. A drawing has a 50° angle. A scale copy is made at k = 2. What is the matching angle in the copy? *1 pt*

- ○ A. 100°
- ○ B. 25°
- ● **C. 50°**
- ○ D. 52°

C

Lengths scale; angles do not.

Scoring: 1 point.

4. A 2 cm by 5 cm rectangle is scaled by k = 4.

a. Find the copy's perimeter. *1 pt*

56 cm

perimeter × k

b. Find the copy's area. *1 pt*

160 cm²

area × k² — 36, 40 and 30 multiplied the area by k once

Scoring: a 1 · b 1 = 2 points.

5. A figure has area 5 cm². Its scale copy has area 80 cm². Find k, and explain why k is not 16. *1 pt*

k = 4. The area ratio is k², because area is two lengths each multiplied by k; 16 is k², so k is its square root.

The point needs k AND the reason (two lengths, so the ratio is k²). 'k = 3' alone is half the item; 'because you square root it' with no reason why is the rule without the reason.

Scoring: 1 point.

6. A floor plan is drawn at ½ in : 1 ft. A room is 3 in by 5 in on the plan.

a. Find the real length of the wall drawn 3 in long. *1 pt*

6 ft

the scale per 1 inch, then × — 1 in : 4 ft, 1 in : 2 ft, 1 in : 5 ft

b. Find the room's real area. *1 pt*

60 ft²

real dimensions first: 8 by 14, 6 by 10, 10 by 15 — the drawing's area × the rate once is the named error

Scoring: a 1 · b 1 = 2 points.

7. A plan is drawn at ¼ in : 1 ft. A real wall is 22 ft long. How long is it on the plan? *1 pt*

5.5 in

real ÷ rate

Scoring: 1 point.

8. A plan is drawn at 1 in : 4 ft. A room is 1.5 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square metres. *1 pt*

About 4.47 m^2 (1.83 m by 2.44 m)

6 ft by 8 ft; 1.83 × 2.44 = 4.4652. The same number by the other route: 48 ft^2 × 0.305^2 = 4.4652. 14.64 (48 × 0.305) used the ratio once on an area. A plan's area asked in metric appears on no review and in no question bank; its type was 5.10 board 8.

Scoring: 1 point.

Part 2. Converting Across Systems *8 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimetres is 11 in? *1 pt*

27.94 cm

× the ratio whose 1 is inches

Scoring: 1 point.

10. 1 cm = 0.39 in. How many inches is 90 cm? *1 pt*

35.1 in (35.4 by ÷ 2.54)

× 0.39; the sheet's other ratio, ÷ 2.54, earns the point too when the work shows it. × 2.54 (177.8, 228.6, 279.4) is the inverted ratio.

Scoring: 1 point.

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimetres is 5 yd? *1 pt*

457.2 cm

a chain: 144 in, 180 in, 54 in, then × 2.54. The estimate (a yard is nearly a metre) catches 10.16, 12.7 and 3.81.

Scoring: 1 point.

12. 1 lb = 0.454 kg. How many kilograms is 60 lb? *1 pt*

27.24 kg

× the ratio whose 1 is the unit you have; the estimate says more pounds than kilograms

Scoring: 1 point.

13. 1 gal = 3.785 L. How many litres is 9 gal? *1 pt*

34.065 L

more litres than gallons

Scoring: 1 point.

14. 1 ft = 12 in. A tile is 7 ft^2 in area. How many square inches is that? *1 pt*

- ● **A. 1,008 in^2**
- ○ B. 84 in^2
- ○ C. 144 in^2
- ○ D. 168 in^2

A

Scoring: 1 point.

15. Rate on 5 Nov: 1 US dollar = 1.35 Canadian dollars.

a. How many Canadian dollars is $70? *1 pt*

94.50 Canadian dollars

× the rate

b. How many US dollars is 54 Canadian dollars? *1 pt*

$40

÷ the rate; 21.16, 72.90 and 87.12 multiplied both ways

Scoring: a 1 · b 1 = 2 points.

Part 3. Multi-Step Proportion Problems *6 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car does 22 miles per gallon and its tank holds 20 gal.

a. How far does it go on a full tank? *1 pt*

440 mi

the rate × the gallons

b. It starts a 1,500-mi trip with a full tank. How many tanks of gas does the trip take? *1 pt*

4 tanks (1,500 ÷ 440 ≈ 3.41)

the part of a tank is still a tank; follow-through on part (a)

Scoring: a 1 · b 1 = 2 points.

17. A recipe for 6 servings uses 2 cups of rice.

a. How much rice for 9 servings? *1 pt*

3 cups

k from the yields (2.5, 1.5, 3), then the ingredient

b. With 8 cups on hand, how many servings can be made? *1 pt*

24 servings

recipes the pantry makes (4, 4, 5) × the yield; 15, 24 and 25 scaled the yield by the cups

Scoring: a 1 · b 1 = 2 points.

18. Paint is mixed 1 : 3 : 4. A batch uses 12 gal of the second colour. How many gallons is the whole batch? *1 pt*

32 gal

k from the part you know (3, 4, 4), then the whole (9, 8, 6 parts); 12, 16 and 8 found one other part and stopped

Scoring: 1 point.

19. 1 mi = 1.61 km. A 450-km drive. Truck A averages 55 mi/h; truck B averages 90 km/h. Which truck gets there first, and by about how long? Explain. *1 pt*

Truck B: 55 mi/h = 88.55 km/h < 90 km/h, so 5 h against about 5.08 h — about 5 minutes sooner.

The point needs the speeds in one unit, the choice, and the times (or the gap). 'Truck B, 90 > 55' is right by accident — no credit without the conversion. The surface appears on no review and in no question bank; its type was 5.10 board 9.

Scoring: 1 point.

GRADE 7 ACCELERATED · UNIT 5 · PRACTICE TEST

Multi-Step Proportional Relationships

24 points

Name_____
Date_____
Period_____

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors

10 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 4 cm side of a figure is 12 cm in its scale copy. Find the scale factor. *1 pt*

2. A 12 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

3. A drawing has a 25° angle. A scale copy is made at k = 4. What is the matching angle in the copy? *1 pt*

- ○ A. 100°
- ○ B. 6.25°
- ○ C. 29°
- ○ D. 25°

4. A 3 cm by 5 cm rectangle is scaled by k = 2.

a. Find the copy's perimeter. *1 pt*

b. Find the copy's area. *1 pt*

5. A figure has area 6 cm^2. Its scale copy has area 150 cm^2. Find k, and explain why k is not 25. *1 pt*

6. A floor plan is drawn at 1 in : 5 ft. A room is 2 in by 3 in on the plan.

a. Find the real length of the wall drawn 2 in long. *1 pt*

b. Find the room's real area. *1 pt*

7. A plan is drawn at 1 in : 3 ft. A real wall is 15 ft long. How long is it on the plan? *1 pt*

8. A plan is drawn at 1 in : 6 ft. A room is 1.5 in by 2.5 in on the plan. Find the room's real area. *1 pt*

Part 2. Converting Across Systems *8 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimetres is 7 in? *1 pt*

10. 1 cm = 0.39 in. How many inches is 110 cm? *1 pt*

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimetres is 1.5 yd? *1 pt*

12. 1 kg = 2.204 lb. How many pounds is 35 kg? *1 pt*

13. 1 gal = 3.785 L. How many litres is 3 gal? *1 pt*

14. 1 m = 100 cm. A floor is 5 m^2 in area. How many square centimetres is that? *1 pt*

- ○ A. 500 cm^2
- ○ B. 50,000 cm^2
- ○ C. 10,000 cm^2
- ○ D. 1,000 cm^2

15. Rate on 5 Nov: 1 US dollar = 1.32 Canadian dollars.

a. How many Canadian dollars is $90? *1 pt*

b. How many US dollars is 66 Canadian dollars? *1 pt*

Part 3. Multi-Step Proportion Problems *6 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car does 27 miles per gallon and its tank holds 10 gal.

a. How far does it go on a full tank? *1 pt*

b. It starts a 1,200-mi trip with a full tank. How many tanks of gas does the trip take? *1 pt*

17. A recipe for 5 servings uses 2 cups of rice.

a. How much rice for 15 servings? *1 pt*

b. With 10 cups on hand, how many servings can be made? *1 pt*

18. Paint is mixed 3 : 1 : 2. A batch uses 4 gal of the second colour. How many gallons is the whole batch? *1 pt*

19. 1 mi = 1.61 km. A train does 80 km/h; a bus does 45 mi/h. Which is faster? Show both speeds in one unit. *1 pt*

GRADE 7 ACCELERATED · UNIT 5 · PRACTICE TEST

Multi-Step Proportional Relationships

24 points **ANSWER KEY**

Name
Date
Period

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors

10 points MA.7.GR.1.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 4 cm side of a figure is 12 cm in its scale copy. Find the scale factor. *1 pt*

k = 3

copy ÷ original

Scoring: 1 point.

2. A 12 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

k = 0.5

copy ÷ original; the copy is smaller, so k is under 1 — 2.5, 4 and 2 are the proportion backwards

Scoring: 1 point.

3. A drawing has a 25° angle. A scale copy is made at k = 4. What is the matching angle in the copy? *1 pt*

- ○ A. 100°
- ○ B. 6.25°
- ○ C. 29°
- ● **D. 25°**

D

Lengths scale; angles do not.

Scoring: 1 point.

4. A 3 cm by 5 cm rectangle is scaled by k = 2.

a. Find the copy's perimeter. *1 pt*

32 cm

perimeter × k

b. Find the copy's area. *1 pt*

60 cm²

area × k² — 36, 40 and 30 multiplied the area by k once

Scoring: a 1 · b 1 = 2 points.

5. A figure has area 6 cm². Its scale copy has area 150 cm². Find k, and explain why k is not 25. *1 pt*

k = 5. The area ratio is k², because area is two lengths each multiplied by k; 25 is k², so k is its square root.

The point needs k AND the reason (two lengths, so the ratio is k²). 'k = 3' alone is half the item; 'because you square root it' with no reason why is the rule without the reason.

Scoring: 1 point.

6. A floor plan is drawn at 1 in : 5 ft. A room is 2 in by 3 in on the plan.

a. Find the real length of the wall drawn 2 in long. *1 pt*

10 ft

the scale per 1 inch, then × — 1 in : 4 ft, 1 in : 2 ft, 1 in : 5 ft

b. Find the room's real area. *1 pt*

150 ft²

real dimensions first: 8 by 14, 6 by 10, 10 by 15 — the drawing's area × the rate once is the named error

Scoring: a 1 · b 1 = 2 points.

7. A plan is drawn at 1 in : 3 ft. A real wall is 15 ft long. How long is it on the plan? *1 pt*

5 in

real ÷ rate

Scoring: 1 point.

8. A plan is drawn at 1 in : 6 ft. A room is 1.5 in by 2.5 in on the plan. Find the room's real area. *1 pt*

135 ft²

9 ft by 15 ft.

Scoring: 1 point.

Part 2. Converting Across Systems

8 points *MA.7.AR.3.3*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimetres is 7 in? *1 pt*

17.78 cm

× the ratio whose 1 is inches

Scoring: 1 point.

10. 1 cm = 0.39 in. How many inches is 110 cm? *1 pt*

42.9 in (43.3 by ÷ 2.54)

× 0.39; the sheet's other ratio, ÷ 2.54, earns the point too when the work shows it. × 2.54 (177.8, 228.6, 279.4) is the inverted ratio.

Scoring: 1 point.

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimetres is 1.5 yd? *1 pt*

137.16 cm

a chain: 144 in, 180 in, 54 in, then × 2.54. The estimate (a yard is nearly a metre) catches 10.16, 12.7 and 3.81.

Scoring: 1 point.

12. 1 kg = 2.204 lb. How many pounds is 35 kg? *1 pt*

77.14 lb

× the ratio whose 1 is the unit you have; the estimate says more pounds than kilograms

Scoring: 1 point.

13. 1 gal = 3.785 L. How many litres is 3 gal? *1 pt*

11.355 L

more litres than gallons

Scoring: 1 point.

14. 1 m = 100 cm. A floor is 5 m^2 in area. How many square centimetres is that? *1 pt*

- ○ A. 500 cm^2
- ● **B. 50,000 cm^2**
- ○ C. 10,000 cm^2
- ○ D. 1,000 cm^2

B

Scoring: 1 point.

15. Rate on 5 Nov: 1 US dollar = 1.32 Canadian dollars.

a. How many Canadian dollars is $90? *1 pt*

118.80 Canadian dollars

× the rate

b. How many US dollars is 66 Canadian dollars? *1 pt*

$50

÷ the rate; 21.16, 72.90 and 87.12 multiplied both ways

Scoring: a 1 · b 1 = 2 points.

Part 3. Multi-Step Proportion Problems

6 points MA.7.AR.4.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car does 27 miles per gallon and its tank holds 10 gal.

a. How far does it go on a full tank? *1 pt*

270 mi

the rate × the gallons

b. It starts a 1,200-mi trip with a full tank. How many tanks of gas does the trip take? *1 pt*

5 tanks (1,200 ÷ 270 ≈ 4.44)

the part of a tank is still a tank; follow-through on part (a)

Scoring: a 1 · b 1 = 2 points.

17. A recipe for 5 servings uses 2 cups of rice.

a. How much rice for 15 servings? *1 pt*

6 cups

k from the yields (2.5, 1.5, 3), then the ingredient

b. With 10 cups on hand, how many servings can be made? *1 pt*

25 servings

recipes the pantry makes (4, 4, 5) × the yield; 15, 24 and 25 scaled the yield by the cups

Scoring: a 1 · b 1 = 2 points.

18. Paint is mixed 3 : 1 : 2. A batch uses 4 gal of the second colour. How many gallons is the whole batch? *1 pt*

24 gal

k from the part you know (3, 4, 4), then the whole (9, 8, 6 parts); 12, 16 and 8 found one other part and stopped

Scoring: 1 point.

19. 1 mi = 1.61 km. A train does 80 km/h; a bus does 45 mi/h. Which is faster? Show both speeds in one unit. *1 pt*

The train: 45 mi/h = 72.45 km/h, and 80 > 72.45.

Same unit first; 80 > 45 compared two units.

Scoring: 1 point.

Score Tracker

Benchmark	Questions	Possible	Earned
MA.7.GR.1.5	1, 2, 3, 4, 5, 6, 7, 8	10	
MA.7.AR.3.3	9, 10, 11, 12, 13, 14, 15	8	
MA.7.AR.4.5	16, 17, 18, 19	6	
Total		**24**	

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.

GRADE 7 ACCELERATED · UNIT 5 · PRACTICE TEST

Multi-Step Proportional Relationships

24 points **WORKED ANSWERS**

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors *10 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 4 cm side of a figure is 12 cm in its scale copy. Find the scale factor. *1 pt*

k = 3

copy ÷ original

Scoring: 1 point.

2. A 12 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

k = 0.5

copy ÷ original; the copy is smaller, so k is under 1 — 2.5, 4 and 2 are the proportion backwards

Scoring: 1 point.

3. A drawing has a 25° angle. A scale copy is made at k = 4. What is the matching angle in the copy? *1 pt*

- ○ A. 100°
- ○ B. 6.25°
- ○ C. 29°
- ● **D. 25°**

D

Lengths scale; angles do not.

Scoring: 1 point.

4. A 3 cm by 5 cm rectangle is scaled by k = 2.

a. Find the copy's perimeter. *1 pt*

32 cm

perimeter × k

b. Find the copy's area. *1 pt*

60 cm²

area × k² — 36, 40 and 30 multiplied the area by k once

Scoring: a 1 · b 1 = 2 points.

5. A figure has area 6 cm². Its scale copy has area 150 cm². Find k, and explain why k is not 25. *1 pt*

k = 5. The area ratio is k², because area is two lengths each multiplied by k; 25 is k², so k is its square root.

The point needs k AND the reason (two lengths, so the ratio is k²). 'k = 3' alone is half the item; 'because you square root it' with no reason why is the rule without the reason.

Scoring: 1 point.

6. A floor plan is drawn at 1 in : 5 ft. A room is 2 in by 3 in on the plan.

a. Find the real length of the wall drawn 2 in long. *1 pt*

10 ft

the scale per 1 inch, then × — 1 in : 4 ft, 1 in : 2 ft, 1 in : 5 ft

b. Find the room's real area. *1 pt*

150 ft²

real dimensions first: 8 by 14, 6 by 10, 10 by 15 — the drawing's area × the rate once is the named error

Scoring: a 1 · b 1 = 2 points.

7. A plan is drawn at 1 in : 3 ft. A real wall is 15 ft long. How long is it on the plan? *1 pt*

5 in

real ÷ rate

Scoring: 1 point.

8. A plan is drawn at 1 in : 6 ft. A room is 1.5 in by 2.5 in on the plan. Find the room's real area. *1 pt*

135 ft²

9 ft by 15 ft.

Scoring: 1 point.

Part 2. Converting Across Systems *8 points*

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